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import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from scipy import signal
import os

# --- CONFIGURATION ---
folder_path = "normal files"
target_freqs = [50, 100, 150, 200, 250, 300, 350, 400, 450, 500]
csv_files = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

def process_signal(dataframe):
    # Time and Sampling calculation
    t = dataframe['Time'].values
    sig = dataframe['Voltage'].values
    dt = np.mean(np.diff(t))
    fs = 1.0 / dt

    # Remove DC Offset for accurate FFT/PSD
    sig_detrended = sig - np.mean(sig)
    n = len(sig_detrended)

    # 1. FFT Magnitude Calculation
    fft_vals = np.fft.rfft(sig_detrended)
    fft_mag = np.abs(fft_vals) * (2.0 / n)
    freqs_fft = np.fft.rfftfreq(n, d=dt)

    # 2. PSD Calculation (Genuine Periodogram)
    freqs_psd, psd_vals = signal.periodogram(sig_detrended, fs, window='boxcar', detrend=False)

    return freqs_fft, fft_mag, freqs_psd, psd_vals, fs

def print_analysis(file_name, f_fft, m_fft, f_psd, p_psd):
    total_mag = np.sum(m_fft)
    total_power = np.sum(p_psd)

    print(f"\n{'=' * 80}")
    print(f"ANALYSIS FOR: {file_name}")
    print(f"{'Target':<8} | {'Actual':<8} | {'Mag (V)':<10} | {'Mag %':<10} | {'PSD (V^2/Hz)':<15} | {'PSD %':<10}")
    print(f"{'-' * 80}")

    for t in target_freqs:
        # FFT Index
        idx_f = np.abs(f_fft - t).argmin()
        # PSD Index
        idx_p = np.abs(f_psd - t).argmin()

        mag = m_fft[idx_f]
        m_contrib = (mag / total_mag) * 100

        pwr = p_psd[idx_p]
        p_contrib = (pwr / total_power) * 100

        print(f"{'t':<8} | {'f_fft[idx_f]:<8.1f} | {'mag:<10.4f} | {'m_contrib:<10.2f} | {'pwr:<15.2e} | {'p_contrib:<10.2f}")

# --- EXECUTION LOOP ---
for file_name in csv_files:
    file_path = os.path.join(folder_path, file_name)
    try:
        # Load (Assumes 26 line header based on previous code)
        df = pd.read_csv(file_path, skiprows=26, header=None, usecols=[0, 1])
        df.columns = ['Time', 'Voltage']

        # Analysis
        f_fft, m_fft, f_psd, p_psd, fs = process_signal(df)
        print_analysis(file_name, f_fft, m_fft, f_psd, p_psd)

        # Plotting
        fig, (ax1, ax2, ax3) = plt.subplots(3, 1, figsize=(12, 12), layout="constrained")

        # 1. Waveform
        ax1.plot(df['Time'], df['Voltage'], color='black', linewidth=0.8)
        ax1.set_title(f"TIME DOMAIN: {file_name} (Fs: {fs:.1f} Hz)")
        ax1.grid(True, alpha=0.3)

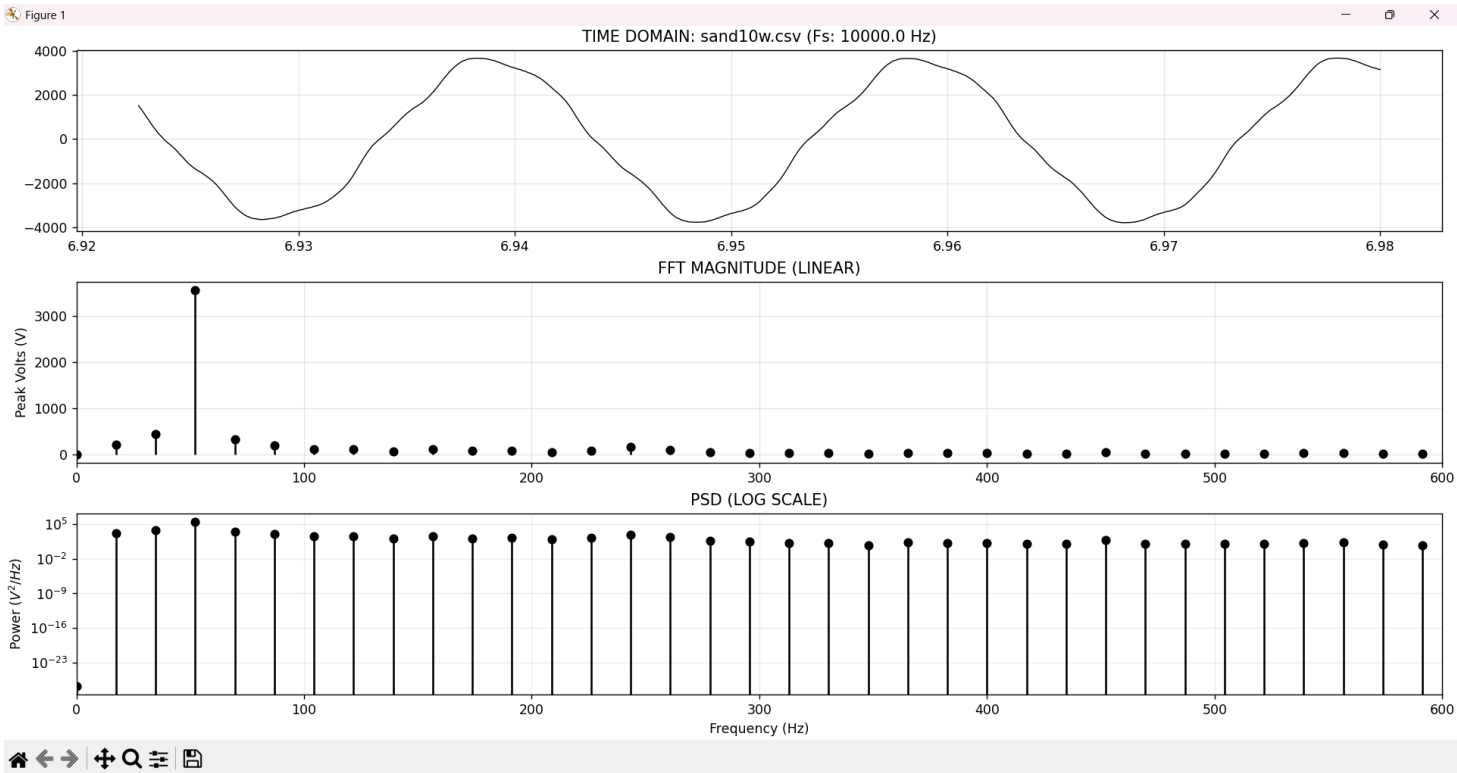
        # 2. FFT Magnitude
        ax2.stem(f_fft, m_fft, linefmt='k-', markerfmt='ko', basefmt=' ')
        ax2.set_xlim(0, 600)
        ax2.set_title("FFT MAGNITUDE (LINEAR)")
        ax2.set_ylabel("Peak Volts (V)")
        ax2.grid(True, alpha=0.3)

        # 3. PSD
        ax3.stem(f_psd, p_psd, linefmt='k-', markerfmt='ko', basefmt=' ')
        ax3.set_xlim(0, 600)
        ax3.set_ylabel("Power (V^2/Hz)")
        ax3.set_title("PSD (LOG SCALE)")
        ax3.grid(True, which="both", alpha=0.2)

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plt.show()
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except Exception as e:  
    print(f"Error processing {file_name}: {e}")
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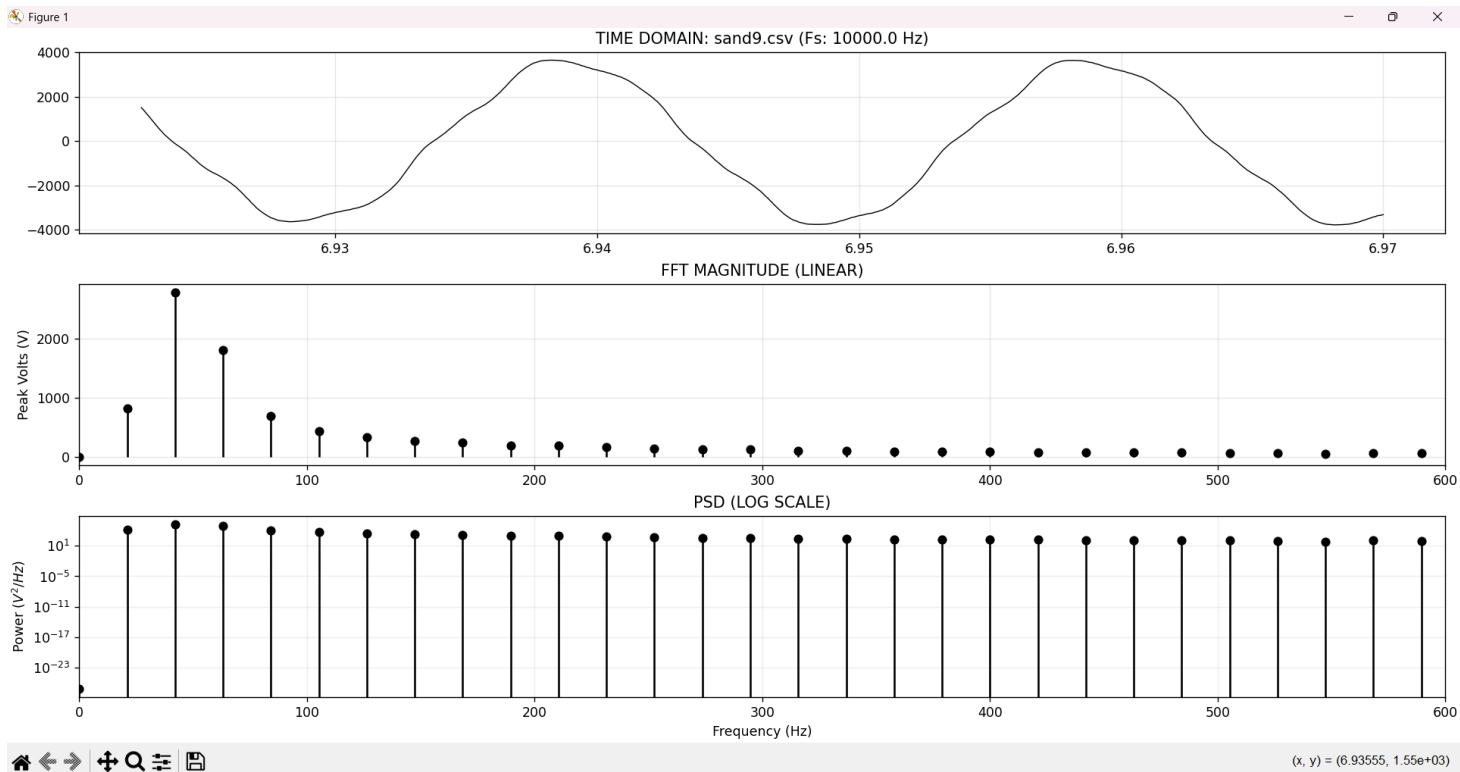


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ANALYSIS FOR: sand10w.csv

Target	Actual	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %

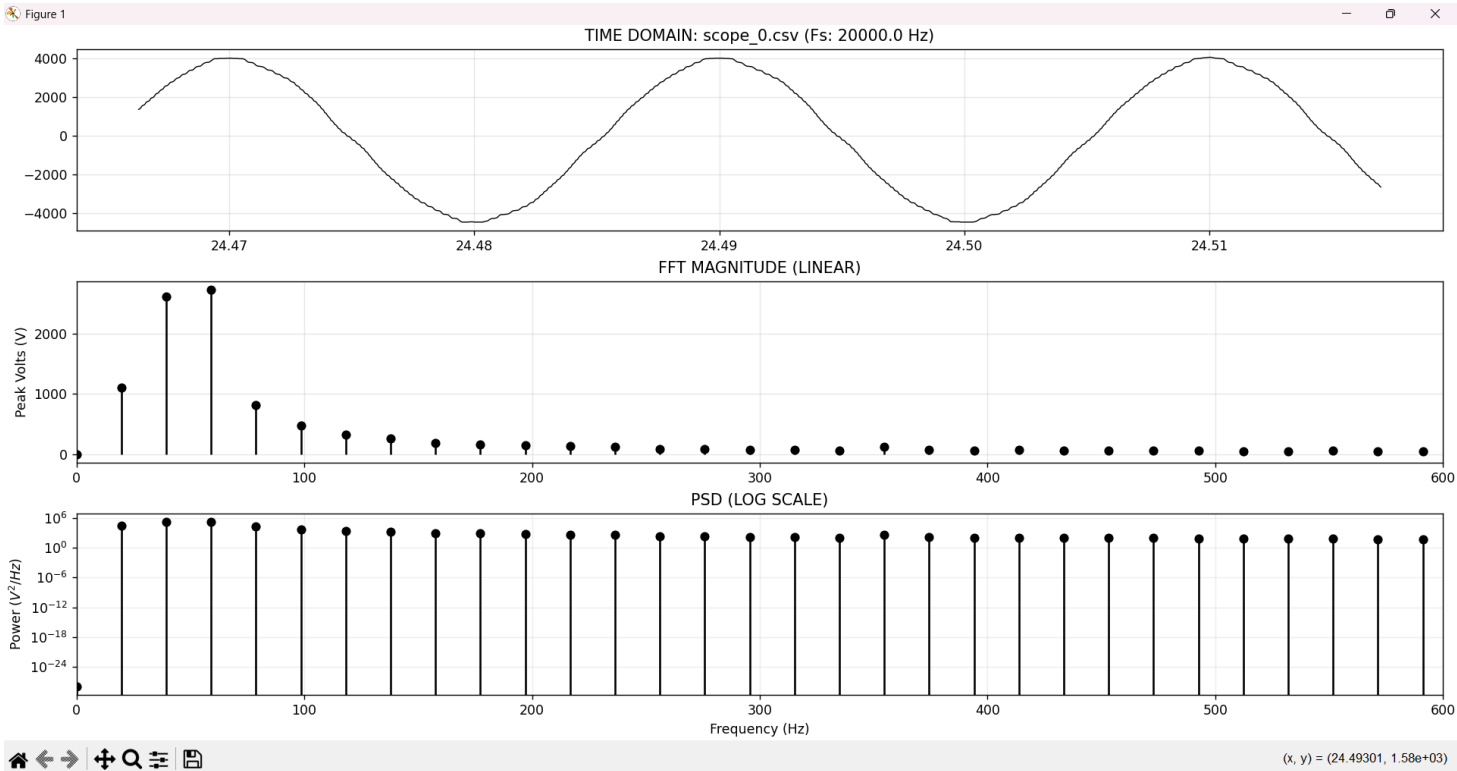
50	52.2	3561.6153	48.80	3.65e+05	96.17
100	104.3	113.3695	1.55	3.70e+02	0.10
150	156.5	117.9009	1.62	4.00e+02	0.11
200	208.7	54.1857	0.74	8.44e+01	0.02
250	243.5	157.0741	2.15	7.09e+02	0.19
300	295.7	33.5854	0.46	3.24e+01	0.01
350	347.8	14.0682	0.19	5.69e+00	0.00
400	400.0	23.5280	0.32	1.59e+01	0.00
450	452.2	51.3716	0.70	7.59e+01	0.02
500	504.3	18.7378	0.26	1.01e+01	0.00



ANALYSIS FOR: sand9.csv

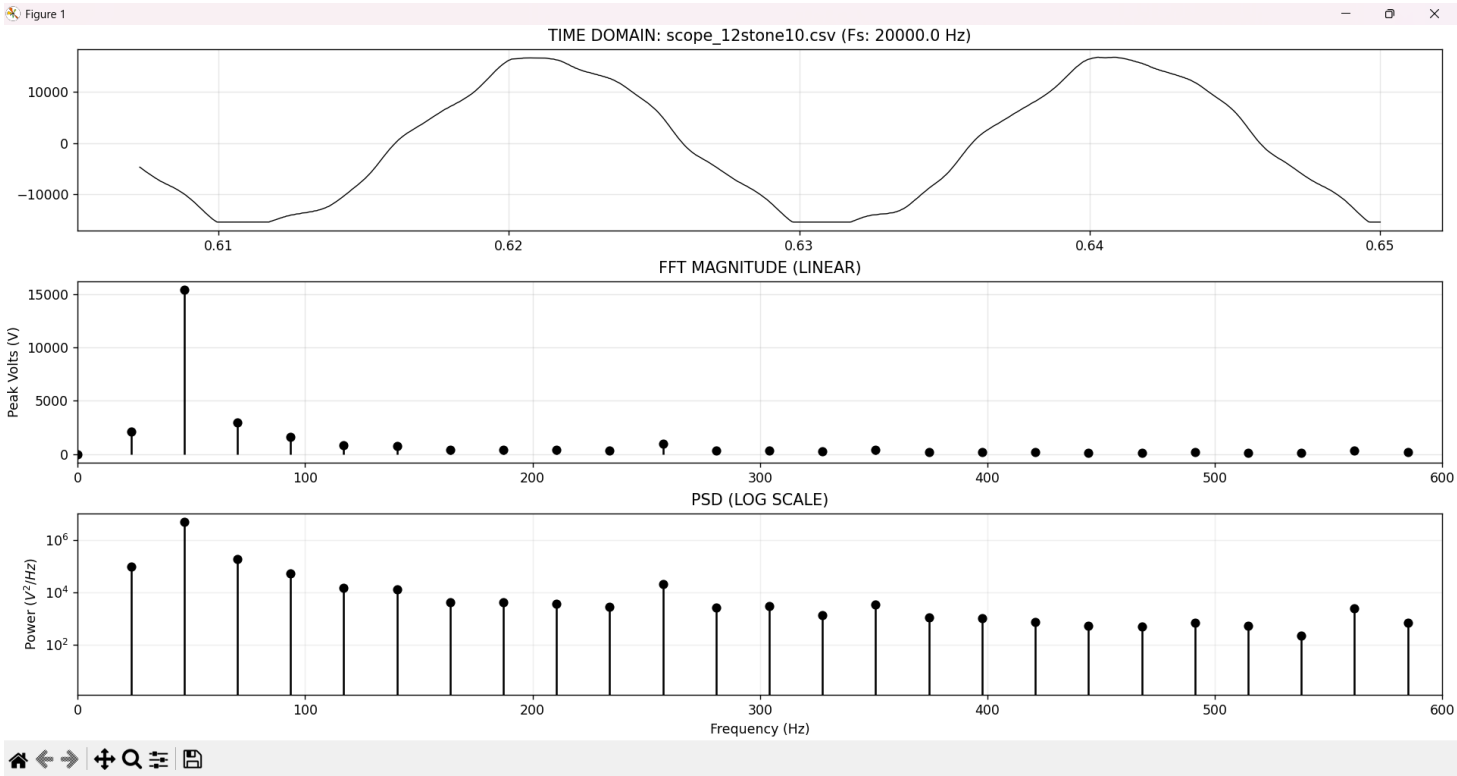
Target	Actual	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
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50	42.1	2790.7455	21.38	1.85e+05	60.12
100	105.3	438.2989	3.36	4.56e+03	1.48
150	147.4	266.9769	2.04	1.69e+03	0.55
200	189.5	190.2187	1.46	8.59e+02	0.28
250	252.6	143.3725	1.10	4.88e+02	0.16
300	294.7	123.9488	0.95	3.65e+02	0.12
350	357.9	90.0326	0.69	1.93e+02	0.06
400	400.0	84.2312	0.65	1.69e+02	0.05
450	442.1	74.0674	0.57	1.30e+02	0.04
500	505.3	66.8016	0.51	1.06e+02	0.03



ANALYSIS FOR: scope_0.csv

Target	Actual	Mag (V)	Mag %	PSD (V ² /Hz)	PSD %
50	59.1	2738.0958	19.27	1.90e+05	44.43
100	98.5	472.8630	3.33	5.67e+03	1.33
150	157.6	182.2616	1.28	8.43e+02	0.20
200	197.0	150.8510	1.06	5.77e+02	0.13
250	256.2	83.1341	0.59	1.75e+02	0.04
300	295.6	78.8146	0.55	1.58e+02	0.04
350	354.7	124.2081	0.87	3.91e+02	0.09
400	394.1	63.6822	0.45	1.03e+02	0.02
450	453.2	65.9427	0.46	1.10e+02	0.03
500	492.6	56.1506	0.40	8.00e+01	0.02

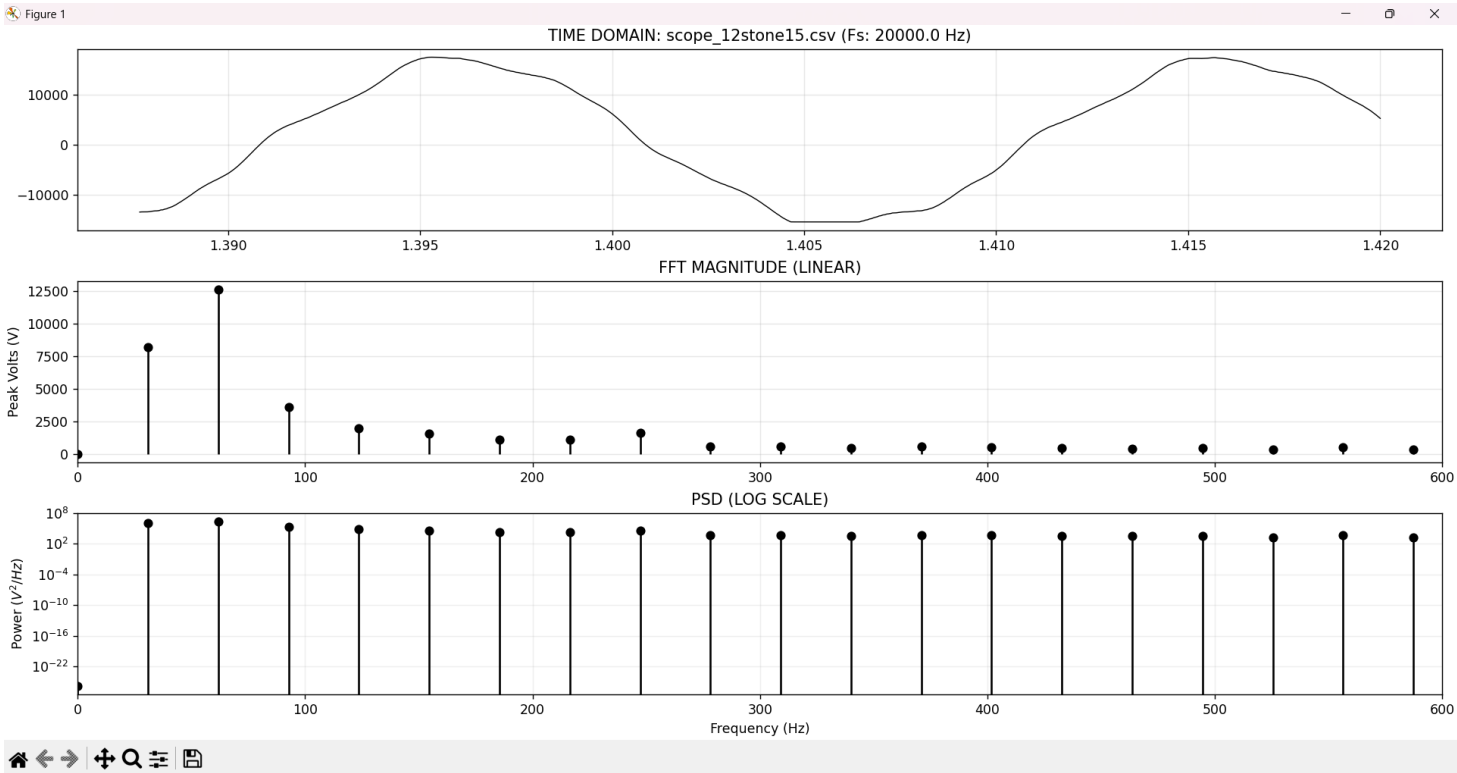


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ANALYSIS FOR: scope_12stone10.csv

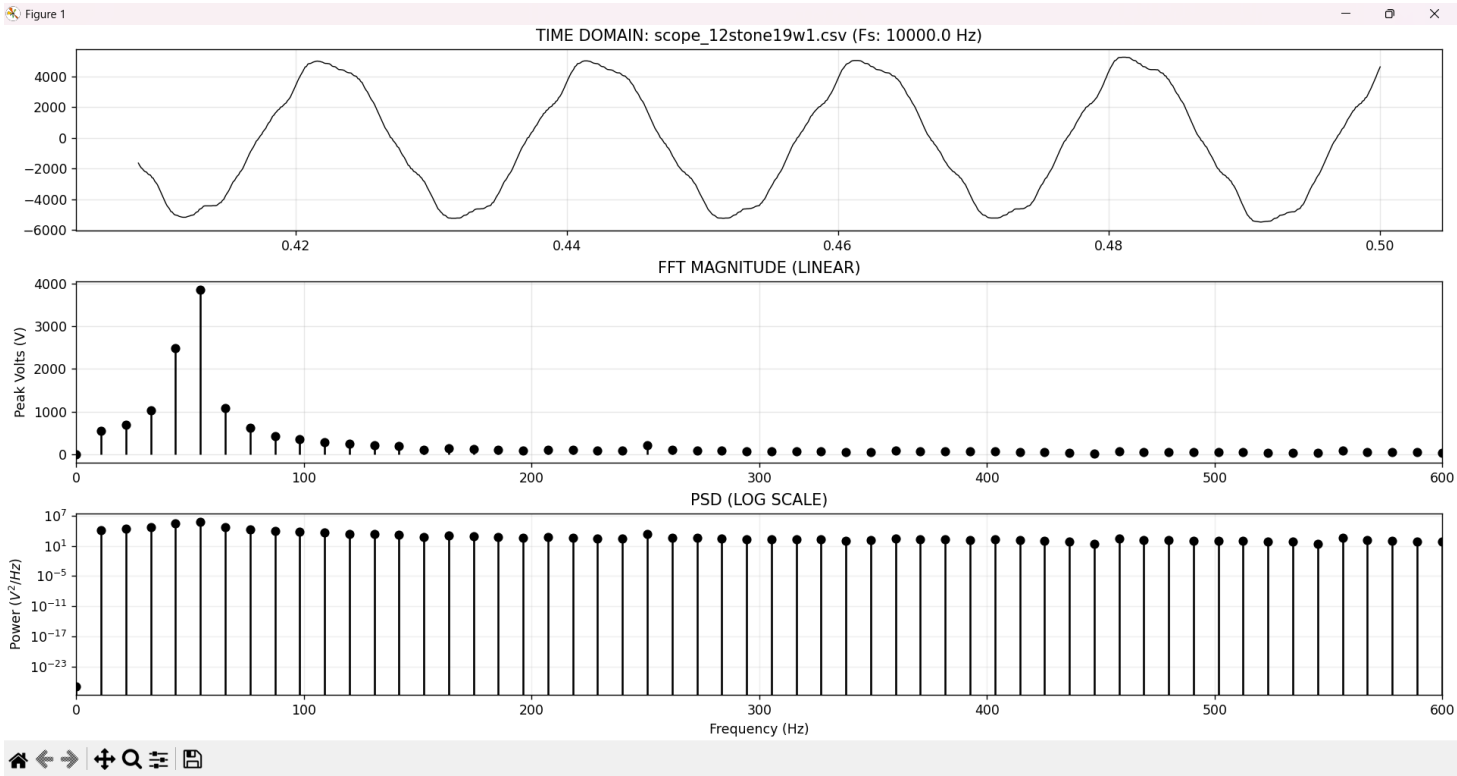
Target | Actual | Mag (V) | Mag % | PSD (V^2/Hz) | PSD %

50	46.8	15431.2273	38.26	5.09e+06	92.25
100	93.6	1586.1438	3.93	5.38e+04	0.97
150	140.4	781.1054	1.94	1.30e+04	0.24
200	210.5	419.2759	1.04	3.76e+03	0.07
250	257.3	976.4831	2.42	2.04e+04	0.37
300	304.1	370.0649	0.92	2.93e+03	0.05
350	350.9	397.4969	0.99	3.38e+03	0.06
400	397.7	217.9593	0.54	1.02e+03	0.02
450	444.4	155.5015	0.39	5.17e+02	0.01
500	491.2	179.8549	0.45	6.91e+02	0.01



ANALYSIS FOR: scope_12stone15.csv

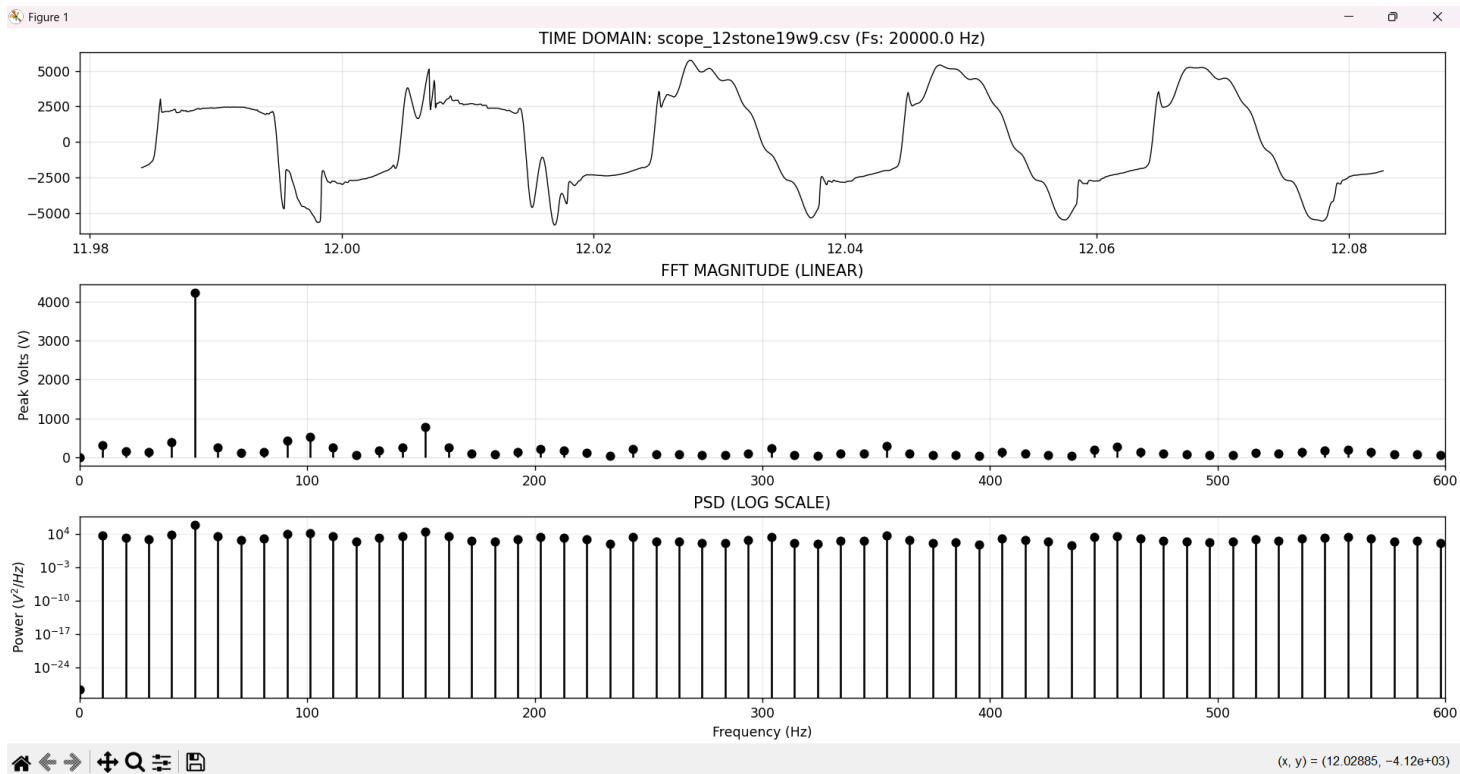
Target	Actual	Mag (V)	Mag %	PSD (V ² /Hz)	PSD %
50	61.8	12654.9932	23.05	2.59e+06	62.75
100	92.7	3583.8013	6.53	2.08e+05	5.03
150	154.6	1535.9035	2.80	3.82e+04	0.92
200	185.5	1097.4525	2.00	1.95e+04	0.47
250	247.3	1595.1014	2.91	4.12e+04	1.00
300	309.1	576.2099	1.05	5.37e+03	0.13
350	340.0	427.0765	0.78	2.95e+03	0.07
400	401.9	501.5848	0.91	4.07e+03	0.10
450	463.7	422.4445	0.77	2.89e+03	0.07
500	494.6	466.2220	0.85	3.52e+03	0.09



ANALYSIS FOR: scope_12stone19w1.csv

Target | Actual | Mag (V) | Mag % | PSD (V^2/Hz) | PSD %

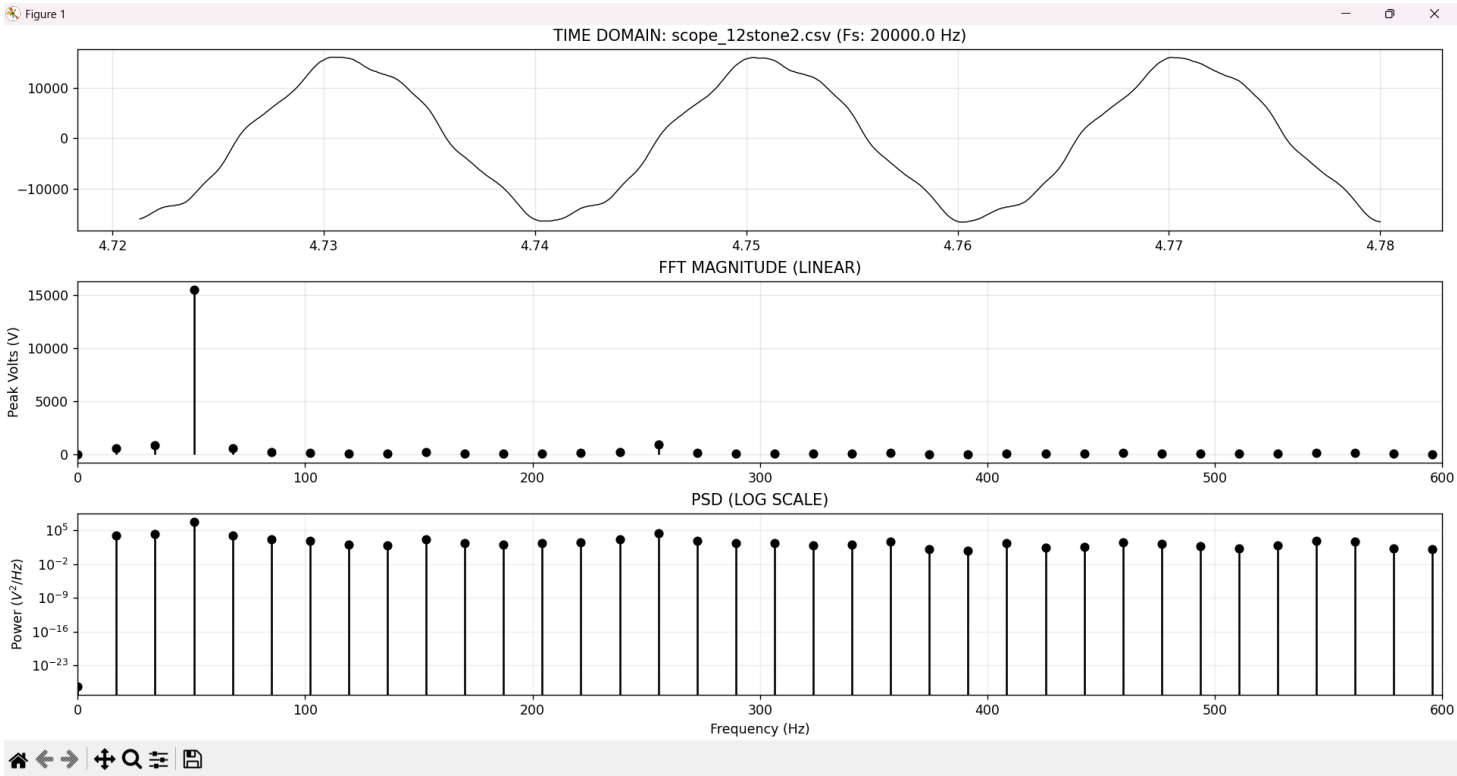
50	54.5	3860.4933	19.48	$6.83e+05$	58.74
100	98.1	355.0199	1.79	$5.78e+03$	0.50
150	152.7	102.6914	0.52	$4.84e+02$	0.04
200	196.3	92.1576	0.47	$3.89e+02$	0.03
250	250.8	215.7118	1.09	$2.13e+03$	0.18
300	305.3	66.6672	0.34	$2.04e+02$	0.02
350	349.0	53.2719	0.27	$1.30e+02$	0.01
400	403.5	65.4245	0.33	$1.96e+02$	0.02
450	447.1	22.9722	0.12	$2.42e+01$	0.00
500	501.6	47.4284	0.24	$1.03e+02$	0.01



ANALYSIS FOR: scope_12stone19w9.csv

Target	Actual	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
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50	50.7	4235.1478	22.07	8.85e+05	87.67
100	101.3	516.1326	2.69	1.31e+04	1.30
150	152.0	773.7816	4.03	2.95e+04	2.93
200	202.6	207.4769	1.08	2.12e+03	0.21
250	253.3	71.6594	0.37	2.53e+02	0.03
300	304.0	222.0253	1.16	2.43e+03	0.24
350	354.6	286.6125	1.49	4.05e+03	0.40
400	395.1	32.6182	0.17	5.25e+01	0.01
450	445.8	197.2668	1.03	1.92e+03	0.19
500	496.5	61.8725	0.32	1.89e+02	0.02

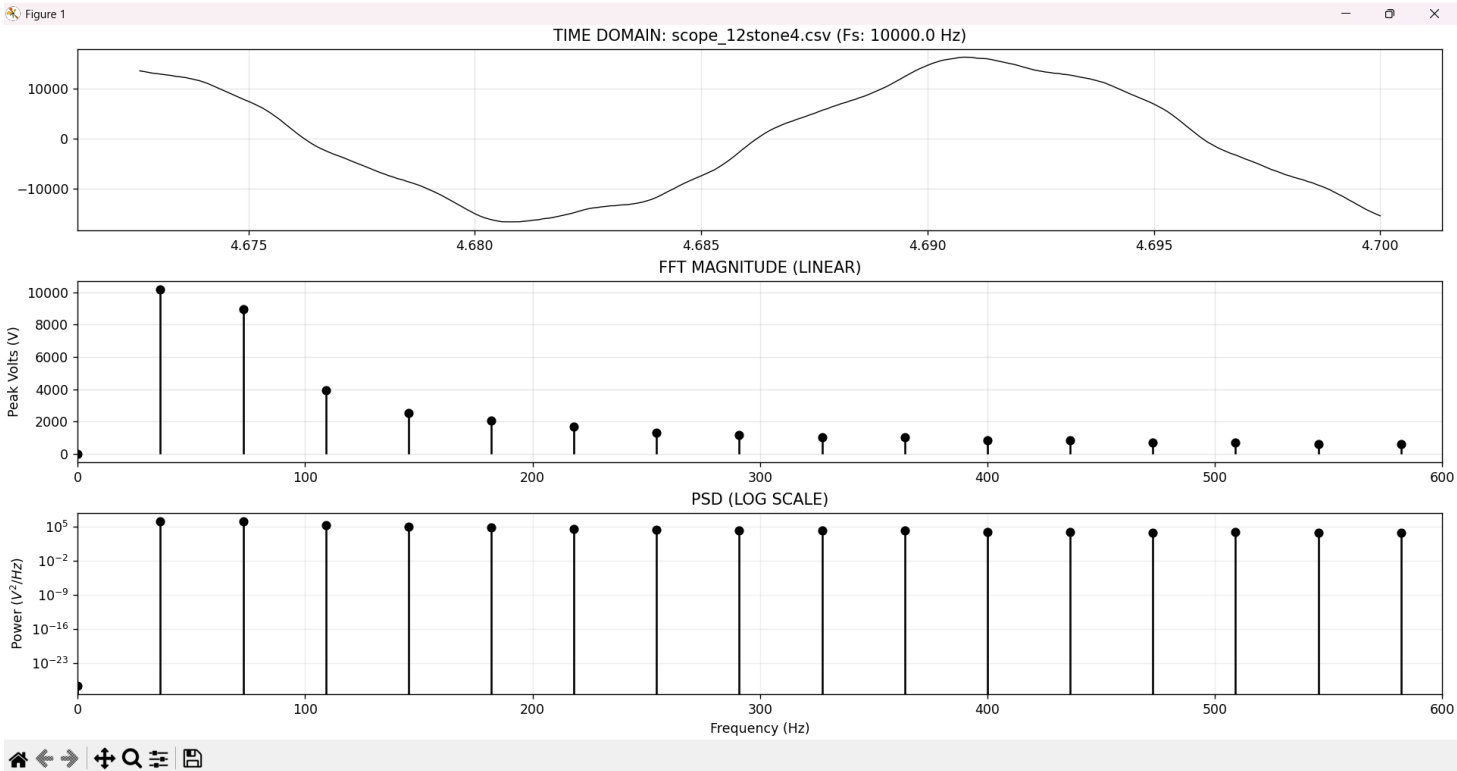


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ANALYSIS FOR: scope_12stone2.csv

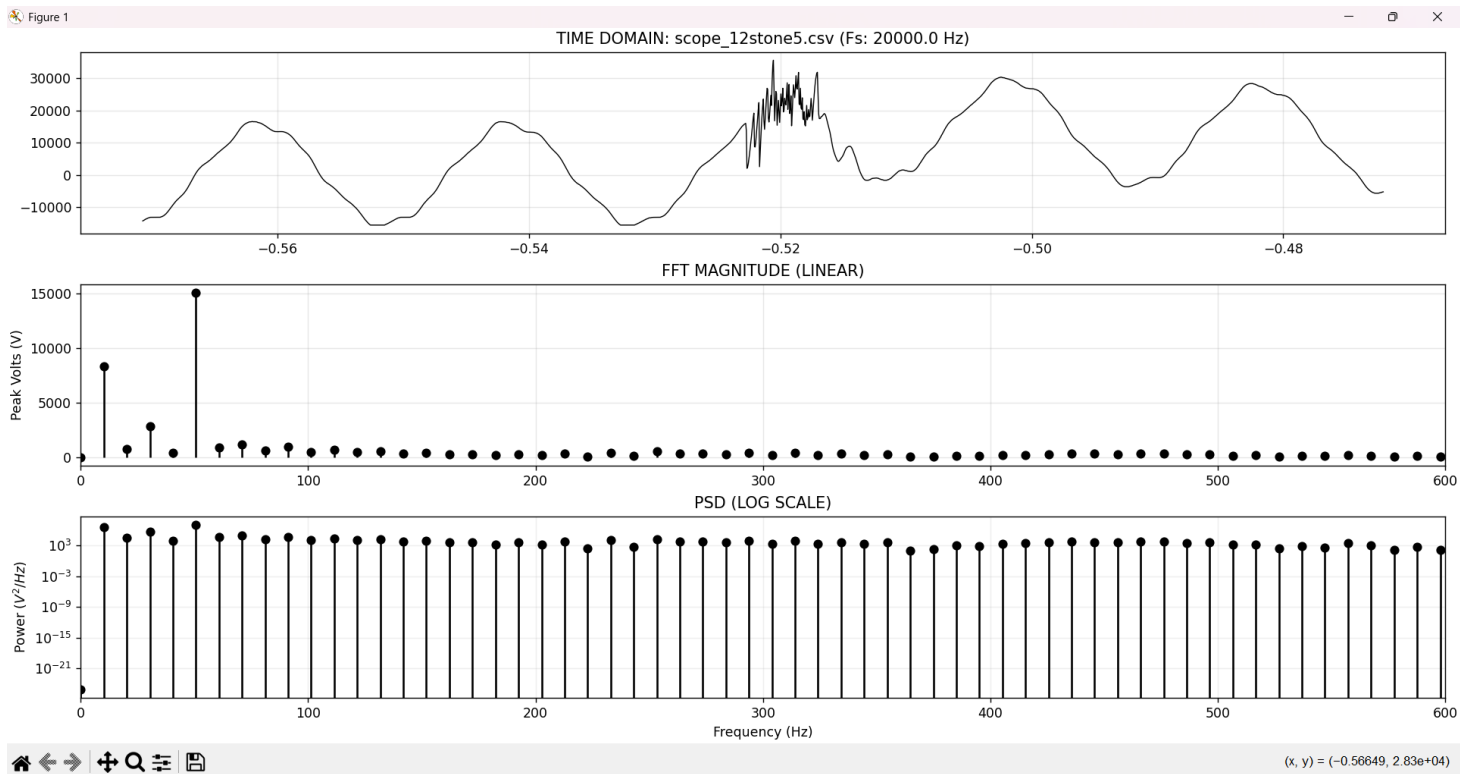
Target	Actual	Mag (V)	Mag %	PSD (V ² /Hz)	PSD %
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50	51.1	15526.2648	70.81	7.08e+06	98.99
100	102.1	141.2986	0.64	5.86e+02	0.01
150	153.2	236.0406	1.08	1.64e+03	0.02
200	204.3	84.0289	0.38	2.07e+02	0.00
250	255.3	914.7194	4.17	2.46e+04	0.34
300	306.4	91.3758	0.42	2.45e+02	0.00
350	357.4	126.5781	0.58	4.71e+02	0.01
400	391.5	13.7087	0.06	5.52e+00	0.00
450	442.6	34.7423	0.16	3.55e+01	0.00
500	493.6	43.1724	0.20	5.48e+01	0.00



ANALYSIS FOR: scope_12stone4.csv

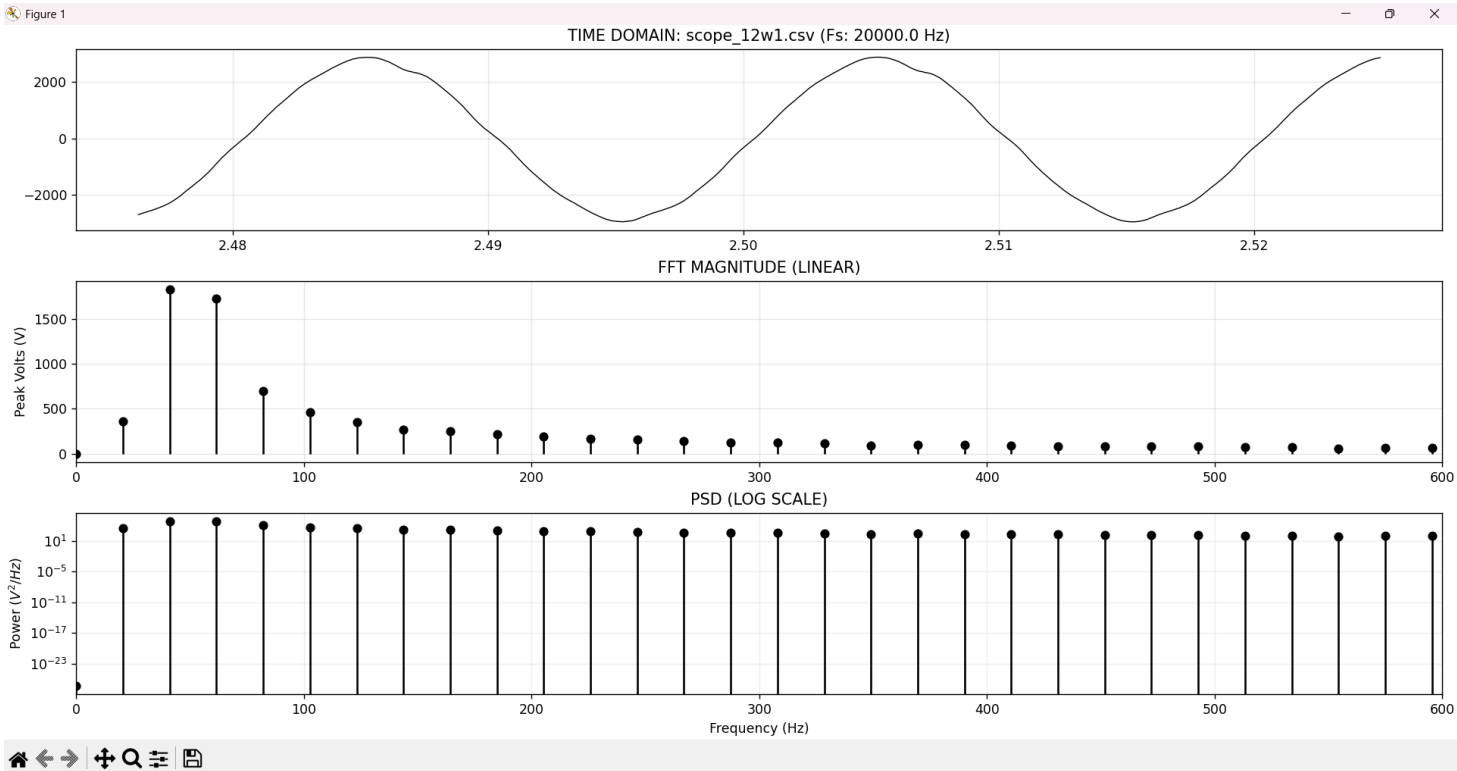
Target	Actual	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
50	36.4	10202.1710	16.94	1.43e+06	45.83
100	109.1	3940.8892	6.54	2.14e+05	6.84
150	145.5	2523.1550	4.19	8.75e+04	2.80
200	218.2	1675.6983	2.78	3.86e+04	1.24
250	254.5	1295.1990	2.15	2.31e+04	0.74
300	290.9	1150.5857	1.91	1.82e+04	0.58
350	363.6	1026.2922	1.70	1.45e+04	0.46
400	400.0	835.1649	1.39	9.59e+03	0.31
450	436.4	819.4424	1.36	9.23e+03	0.30
500	509.1	715.2868	1.19	7.03e+03	0.23



ANALYSIS FOR: scope_12stone5.csv

Target | Actual | Mag (V) | Mag % | PSD (V^2/Hz) | PSD %

50	50.7	15128.7821	18.08	1.13e+07	72.07
100	101.3	452.6222	0.54	1.01e+04	0.06
150	152.0	406.3727	0.49	8.15e+03	0.05
200	202.6	163.6994	0.20	1.32e+03	0.01
250	253.3	561.2805	0.67	1.55e+04	0.10
300	304.0	214.9990	0.26	2.28e+03	0.01
350	354.6	277.1411	0.33	3.79e+03	0.02
400	395.1	118.6201	0.14	6.94e+02	0.00
450	445.8	303.5478	0.36	4.55e+03	0.03
500	496.5	274.7696	0.33	3.73e+03	0.02



ANALYSIS FOR: scope_12w1.csv

Target	Actual	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
50	41.1	1829.4539	13.45	8.15e+04	42.95
100	102.7	463.1730	3.41	5.22e+03	2.75
150	143.7	271.6034	2.00	1.80e+03	0.95
200	205.3	191.6354	1.41	8.94e+02	0.47
250	246.4	160.6709	1.18	6.29e+02	0.33
300	308.0	125.3875	0.92	3.83e+02	0.20
350	349.1	91.3915	0.67	2.03e+02	0.11
400	390.1	97.8973	0.72	2.33e+02	0.12
450	451.7	79.4793	0.58	1.54e+02	0.08
500	492.8	78.7430	0.58	1.51e+02	0.08